Ex.No. 15: AI-Based Traffic Prediction for 5G Networks

AIM:

To design and simulate an **AI-based traffic prediction system for 5G networks using MATLAB** by analyzing historical network traffic data and predicting future traffic demand using a suitable machine learning model.

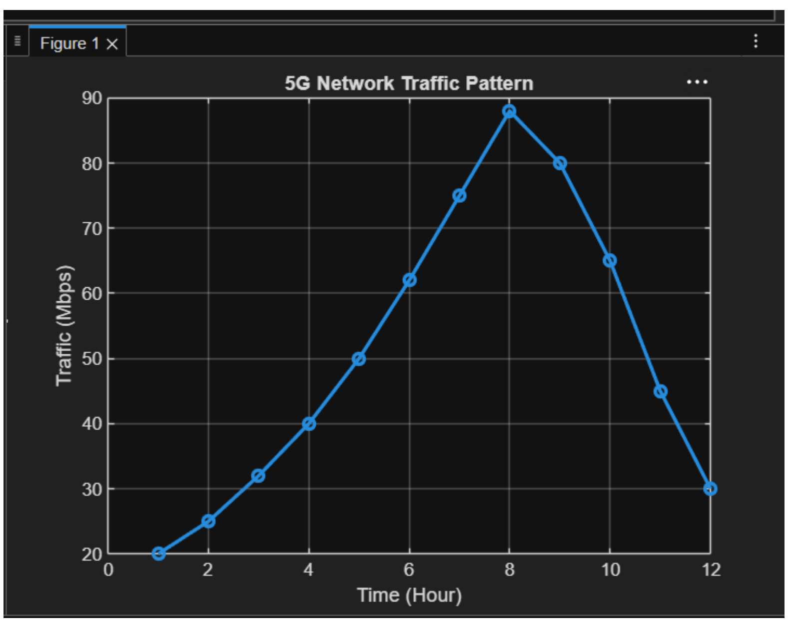
Objective:

1. To analyze **5G network traffic patterns** based on parameters such as time, traffic load, and number of connected users.
2. To develop an **AI-based prediction model** for estimating future 5G network traffic demand.
3. To evaluate the performance of the traffic prediction model and analyze its usefulness for **efficient resource allocation and network management**.

Objective 1: Analyze 5G Network Traffic Patterns

```
clc;

clear;

close all;


% Time (hours)

Time = 1:12;


% Number of connected users

Users = [50 60 75 90 110 130 150 170 160 140 100 80];


% Network Traffic (Mbps)

Traffic = [20 25 32 40 50 62 75 88 80 65 45 30];


% Display data
```

```
disp('Time(h)   Users   Traffic(Mbps)');
disp([Time' Users' Traffic']);

% Plot traffic pattern
figure;

plot(Time, Traffic, '-o', 'LineWidth', 2);

title('5G Network Traffic Pattern');
xlabel('Time (Hour)');
ylabel('Traffic (Mbps)');
grid on;
```

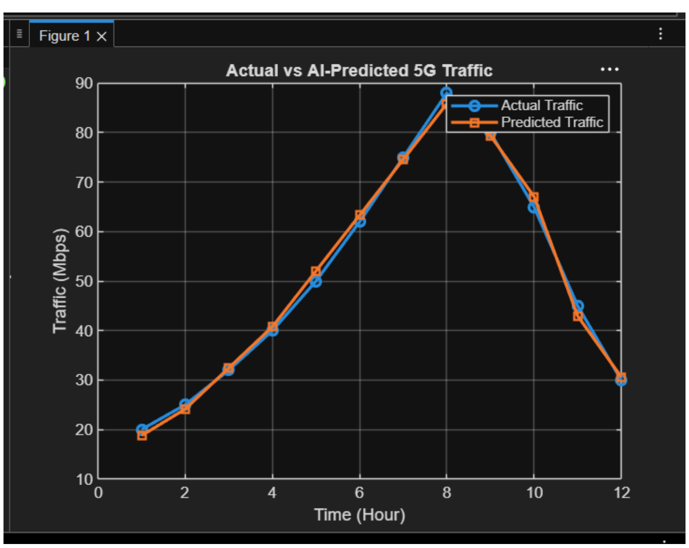


Objective 2: Develop an AI-Based Traffic Prediction Model

```
clc;
clear;
```

```
close all;

% Time (hours)
Time = 1:12;

% Number of connected users
Users = [50 60 75 90 110 130 150 170 160 140 100 80];

% Actual Traffic (Mbps)
Traffic = [20 25 32 40 50 62 75 88 80 65 45 30];

% Input features
X = [Time' Users'];

% Target output
Y = Traffic';

% Train Linear Regression Model
Model = fitlm(X, Y);

% Predict traffic
Predicted_Traffic = predict(Model, X);

% Display results
disp('Time   Actual Traffic   Predicted Traffic');
```

```
disp([Time' Traffic' Predicted_Traffic]);

% Plot actual and predicted traffic
figure;

plot(Time, Traffic, '-o', 'LineWidth', 2);
hold on;

plot(Time, Predicted_Traffic, '-s', 'LineWidth', 2);

title('Actual vs AI-Predicted 5G Traffic');
xlabel('Time (Hour)');
ylabel('Traffic (Mbps)');

legend('Actual Traffic', 'Predicted Traffic');

grid on;
```

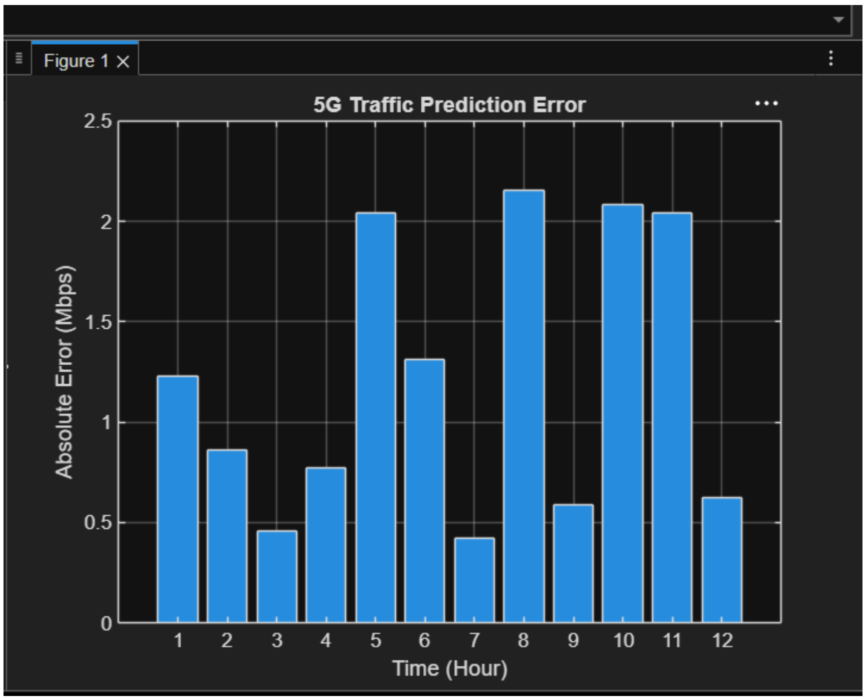


Objective 3: Evaluate AI Traffic Prediction Performance

```
clc;

clear;

close all;


% Time (hours)

Time = 1:12;


% Number of connected users

Users = [50 60 75 90 110 130 150 170 160 140 100 80];


% Actual Traffic (Mbps)

Traffic = [20 25 32 40 50 62 75 88 80 65 45 30];


% Input features

X = [Time' Users'];


% Target output

Y = Traffic';


% Train Linear Regression Model

Model = fitlm(X, Y);


% Predict traffic

Predicted_Traffic = predict(Model, X);
```

```
% Calculate prediction error
Error = Traffic' - Predicted_Traffic;

% Mean Absolute Error
MAE = mean(abs(Error));

% Root Mean Square Error
RMSE = sqrt(mean(Error.^2));

% R-Squared
R2 = 1 - sum(Error.^2) / ...
   sum((Traffic' - mean(Traffic)).^2);

% Display performance
disp('AI Model Performance');

fprintf('Mean Absolute Error (MAE) = %.2f Mbps\n', MAE);
fprintf('Root Mean Square Error (RMSE) = %.2f Mbps\n', RMSE);
fprintf('R-Squared (R2) = %.4f\n', R2);

% Plot prediction error
figure;

bar(Time, abs(Error));

title('5G Traffic Prediction Error');
```

```
xlabel('Time (Hour)');

ylabel('Absolute Error (Mbps)');


grid on;
```



Results

Objective 1: Analyze 5G Network Traffic Patterns

Result:
The 5G network traffic pattern was successfully analyzed using the number of connected users and traffic load. The graph shows that **network traffic increases as the number of connected users increases**, reaching a maximum traffic load of approximately **88 Mbps**.

Result: 5G network traffic variation was successfully simulated and visualized.

Objective 2: Develop an AI-Based Traffic Prediction Model

Result:
A **linear regression-based AI model** was successfully developed to predict 5G network traffic using **time and number of connected users** as input parameters. The actual and predicted traffic values were plotted for comparison.

Result: The AI model successfully predicted the 5G network traffic pattern.

Objective 3: Evaluate AI Traffic Prediction Performance

Result:
The performance of the AI-based traffic prediction model was evaluated using **Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2**. The prediction error was also plotted.

Result: The AI model provided a suitable prediction of network traffic and can be used for **traffic management, resource allocation, and congestion reduction in 5G networks**.